PHYSICAL AI | MODULE 1 REVIEW ASSESSMENT

Lessons 1-5 | Ten-question assessment | Evidence report

First-attempt score	8 / 10 (80%)
Question 7	Corrected after feedback
Question 10	Targeted reinforcement passed
Module 2	Ready; Lesson 6 not started

Evidence basis: User answers and instructor evaluations from the project conversation. This is not a proctored exam. Scores reflect first submissions, not corrected responses. Local Isaac Sim USD files are not attached.

Question-by-question assessment

Q1. Closed-loop control | Correct first attempt

Prompt: A robot gripper detects a cup at 10 cm. During motion, the cup moves to 15 cm. Compare A: continue toward the old target without observing again; B: observe again and adjust. Which is closed-loop control?

Learner response: Strategy B. "Because when the cup position is changed, the whole environment also changes. So, observing the environment again is required."

Evaluation: Strategy B: updated observations inform new actions. Refinement: a change to the cup does not mean the entire environment changes.

Q2. Observation, action, policy | Correct first attempt

Prompt: The camera detects a cup 12 cm away and the controller commands a 2 cm forward motion. Identify the observation, action, and policy.

Learner response: Observation: cup is 12 cm away. Action: move gripper 2 cm. Policy: uses observations to select actions.

Evaluation: Observation = distance measurement; action = movement command; policy = rule mapping observations to actions.

Q3. 3D position updates | Correct first attempt

Prompt: Gripper starts at (2,3,1). It moves 3 units right, 2 backward, and 4 up; +X right, +Y forward, +Z up. Find its final position.

Learner response: (5,1,5). X=2+3; Y=3-2; Z=1+4.

Evaluation: (5,1,5).

Q4. Signed rotation and fixed position | Correct first attempt

Prompt: Gripper position is (4,2,1), orientation +30 deg about Z. It rotates 90 deg clockwise about its own center without translating. Find final orientation and position.

Learner response: Z orientation = -60 deg; position = (4,2,1) unchanged. 30+(-90)=-60.

Evaluation: Orientation -60 deg, position unchanged.

Q5. 2D robot pose | Correct first attempt

Prompt: A mobile robot starts at pose (3,2,45 deg), moves 2 world-X units with world Y unchanged, then rotates 45 deg counterclockwise about its center. Find the final pose.

Learner response: (5,2,90 deg). X=3+2; Y=2+0; "Z"=45+45.

Evaluation: (5,2,90 deg). Notation refinement: the third 2D pose component is theta (heading), not Z position.

Q6. Robot and world reference frames | Correct first attempt

Prompt: Robot world position (2,3); cup world position (8,7). Find cup robot-frame coordinates at yaw 0 deg and after robot yaw +90 deg, with both objects stationary.

Learner response: Before (6,4); after (4,-6). Robot rotation changes local X/Y axes.

Evaluation: Before (6,4) by subtraction; after (4,-6) using inverse rotation.

Q7. Inverse rotation matrix | Incorrect first attempt; corrected on retry

Prompt: Robot at world origin, yaw +90 deg; cube world point (5,2). Compute cube coordinates in the robot frame using R(-90)=[[0,1],[-1,0]].

Learner response: First: (-2,5), with X=0+(-2), Y=5+0. Retry: (2,-5), with X=0+2, Y=-5+0.

Evaluation: (2,-5). x=0*5+1*2=2; y=-1*5+0*2=-5. First attempt applied the positive rather than inverse rotation.

Q8. Combined world-to-robot transform | Correct first attempt

Prompt: Robot world position (2,1), yaw +90 deg; cube world position (6,4). Compute displacement and robot-frame coordinates using inverse rotation.

Learner response: World displacement (4,3); robot-frame (3,-4). X=0+3; Y=-4+0.

Evaluation: Displacement (4,3); inverse-rotated point (3,-4).

Q9. Robot-to-world inverse transform | Correct first attempt

Prompt: Robot world position (3,2), yaw +90 deg; cube robot-frame position (4,-2). Calculate rotated vector and world point.

Learner response: Rotated vector (2,4); world coordinates (5,6).

Evaluation: R(+90)*(4,-2)=(2,4); then +(3,2)=(5,6).

Q10. Parent-child transforms | Incorrect first attempt; later reinforcement passed

Prompt: Isaac Sim: RobotFrame parent translation (3,2,0), yaw +90 deg; child Cube local position (2,1,1). Find rotated XY and world position.

Learner response: First: rotated XY (-3,5); world (0,7,1). Reasoning: "I am not sure."

Evaluation: Rotated XY (-1,2) using (x,y)->(-y,x); world position (2,4,1) after adding translation (3,2,0). Z=1 was correct.

Follow-up reinforcement after Question 10

New task: Parent translation (4,1,0), yaw +90 degrees, child local position (3,2,1). The learner calculated rotated XY (-2,3) and world position (2,4,1). Both were correct. This was a separate, successful independent reinforcement after feedback.

Final interpretation and recommendation

First attempts: 8 correct, 2 incorrect. Q7 was correctly revised after feedback. Q10 was not re-answered as the original item, but a different targeted problem testing the same parent-child transform was solved correctly. The learner is ready to proceed to Module 2 with continued practice on inverse-rotation signs and composition order. Lesson 6 has not begun.

Core formula reference

Angle change: delta theta = theta_final - theta_initial

World to robot: p_R = R(-theta) (p_W - t)

Robot to world: p_W = R(+theta) p_R + t

At +90 degrees: (x,y) becomes (-y,x); at -90 degrees: (x,y) becomes (y,-x).